

AI Lab # 9

This lab explores fuzzy logic-based reasoning by implementing fuzzification, rule evaluation, and defuzzification to enable smooth, human-like decision-making under uncertainty.

Artificial Intelligence (LAB)

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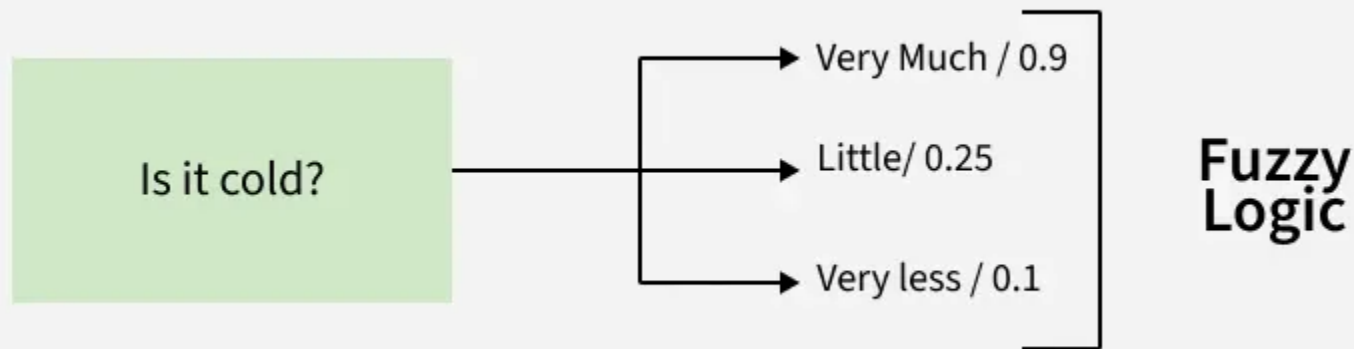
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Today's Agenda

- 01** Crisp vs Fuzzy Thinking
- 02** Fuzzification
- 03** Defuzzification and Its Methods
- 04** Real World Applications
- 05** Example and Lab Tasks

Crisp vs Fuzzy Thinking

- When computers or machines take decisions, they usually work with **crisp values** (clear numbers).
- Example: If temperature $> 30 \rightarrow$ Turn AC ON, else OFF.
But in real life, decisions are **not always black and white**.
- At 28°C , should the AC be ON or OFF? Maybe it should run **a little bit**. This “in-between” thinking is where **fuzzy logic** helps.



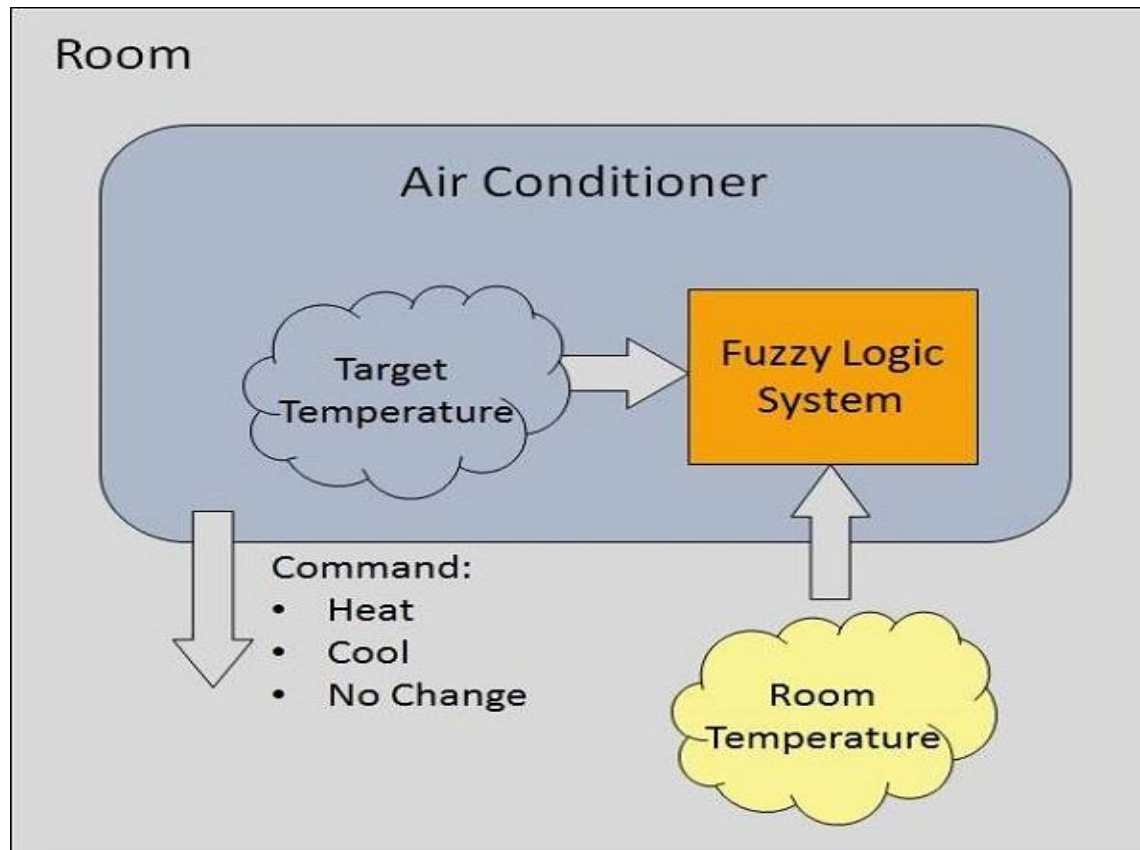
Analogy:

- Think about your mom making **tea**:
- If tea is **too sweet** $\rightarrow$ **add more water**.
- If tea is **not sweet enough** $\rightarrow$ **add sugar**.
- But sometimes it's "**just a little too sweet**" $\rightarrow$ she adds **just a little water**.

That's **fuzzy logic** in real life!

Fuzzification

- Fuzzification means **changing exact input values into fuzzy categories** using membership functions.



Example:

Cold: 0–15°C

Warm: 15–30°C

Hot: 30–45°C

If the temperature = **25°C**

Cold = 0

Warm = 0.8

Hot = 0.2

So, 25°C is mostly “Warm” but a little “Hot.”

Fuzzy Fan Speed

Fan that works on fuzzy logic to adjust fan speed based on the room temperature.



```
def fuzzy_fan_speed(temp):
    if temp <= 15:
        return "SLOW"
    elif 15 < temp <= 25:
        return "MEDIUM"
    else:
        return "FAST"

print(fuzzy_fan_speed(10)) # Expected: SLOW
print(fuzzy_fan_speed(20)) # Expected: MEDIUM
print(fuzzy_fan_speed(30)) # Expected: FAST
```

Defuzzification

- Defuzzification means converting fuzzy outputs into one exact (crisp) value that a machine can actually use.
- It works like taking an **average based on importance**. For example:

Medium fan (60%) has weight 0.8

High fan (90%) has weight 0.2

So, **Fan Speed** = $(0.8 \times 60) + (0.2 \times 90) = 66\%$

This means at 25°C, the AC fan will run at **66% speed**.

So, at 25°C, AC fan runs at **66% speed**.

Why Do We Need It?

- Fuzzy logic gives us answers in **degrees** (0.2, 0.7, etc.).
- Machines need **one crisp value** to work.
- Defuzzification is the **translator** between fuzzy logic and real actions.

Defuzzification Method # 1 - Centroid

- The crisp output is the weighted average of all output values, weighted by their membership degrees — like finding the balance point of a seesaw.
- **Example:**
Warm = 0.3 → Fan = 60%, Hot = 0.7 → Fan = 90%
 $z^* = (0.3 \times 60 + 0.7 \times 90) / (0.3 + 0.7) \approx 75\%$

Code Snippet:

```
cold, warm, hot = 0.0, 0.3, 0.7
```

```
# Crisp values
```

```
cold_val, warm_val, hot_val = 30, 60, 90
```

```
num = (cold * cold_val) + (warm * warm_val)  
+ (hot * hot_val)
```

```
den = cold + warm + hot
```

```
centroid = num / den
```

```
print("Centroid:", centroid)
```

Defuzzification Method # 2 - Maximum Membership

- The crisp output is simply the value corresponding to the rule with the highest membership score — ignore the rest.
- **Example:**
Warm = 0.3, Hot = 0.7 → Hot has the highest score
 $z^* = \text{Fan} = 90\% (\text{Hot})$

Code Snippet:

```
cold, warm, hot = 0.0, 0.3, 0.7
```

```
if hot >= warm and hot >= cold:
```

```
    result = 90
```

```
elif warm >= cold and warm >= hot:
```

```
    result = 60
```

```
else:
```

```
    result = 30
```

```
print("Maximum Membership:", result)
```


Defuzzification Method # 3 - Mean of Maximum

- If two or more rules tie for the highest score, take the average of their corresponding crisp values.
- **Example:**
 Warm = 0.7 → Fan = 60%, Hot = 0.7 → Fan = 90% (tie)
 $z^* = (60 + 90) / 2 = 75\%$

Code Snippet:

```

cold, warm, hot = 0.0, 0.7, 0.7
winners = []
if warm == max(cold, warm, hot):
    winners.append(60)
if hot == max(cold, warm, hot):
    winners.append(90)
if cold == max(cold, warm, hot):
    winners.append(30)

mean_of_max = sum(winners) / len(winners)
print("Mean of Maximum:", mean_of_max)

```

Defuzzification Method # 4 - Smallest of Maximum

- When rules tie for the highest membership, pick the smallest of the tied crisp values — a conservative choice.
- **Example:**
Warm = 0.7 → Fan = 60%, Hot = 0.7 → Fan = 90% (tie)
 $z^* = \min(60, 90) = 60\%$

Code Snippet:

```
cold, warm, hot = 0.0, 0.7, 0.7
winners = []
if warm == max(cold, warm, hot):
    winners.append(60)
if hot == max(cold, warm, hot):
    winners.append(90)

print("Smallest of Maximum:", min(winners))
```

Defuzzification Method # 5 - Largest of Maximum

- When rules tie for the highest membership, pick the largest of the tied crisp values — an aggressive choice.
- **Example:**
Warm = 0.7 → Fan = 60%, Hot = 0.7 → Fan = 90% (tie)
 $z^* = \max(60, 90) = 90\%$

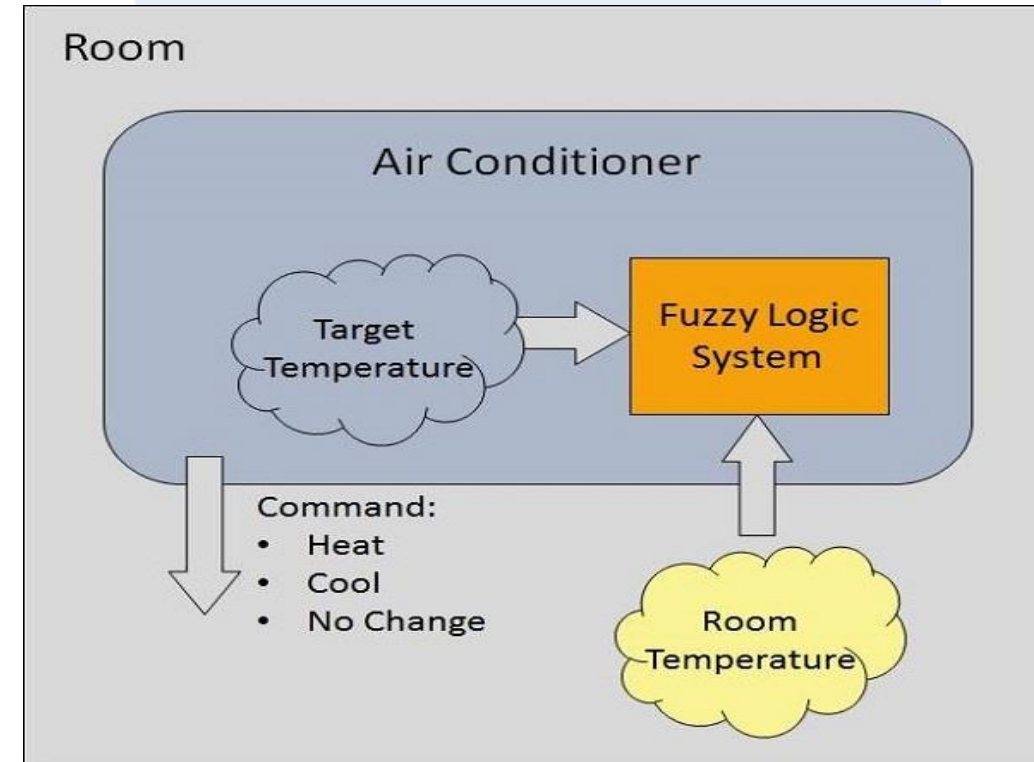
Code Snippet:

```
cold, warm, hot = 0.0, 0.7, 0.7
winners = []
if warm == max(cold, warm, hot):
    winners.append(60)
if hot == max(cold, warm, hot):
    winners.append(90)

print("Largest of Maximum:", max(winners))
```

Real World Applications

- **AC & Refrigerator control systems** (adjust cooling smoothly).
- **Washing machines** (decide wash time based on “little dirty” or “very dirty” clothes).
- **Self-driving cars** (decide actions based on “close,” “far,” or “very close” distances).
- **Medical expert systems** (doctor-like apps saying “mild fever,” “high fever”).
- **Robotics** (smooth movement instead of jerky ON/OFF actions).
- **Recommendation systems** (suggesting movies/music as “maybe like,” “probably like,” “definitely like”).



Lab Task: Air Conditioner Control System (Fuzzification & Defuzzification)

- *Design a fuzzy system that adjusts AC cooling based on temperature and humidity.*
- *Define fuzzy sets for temperature (Cool, Comfortable, Hot) and humidity (Low, High).*
- *Define fuzzy sets for cooling power (Low, Medium, High).*
- *Write rules combining temperature and humidity.*
- *Show output cooling levels for three different input cases.*

Thank You

Questions or discussions are welcomed!